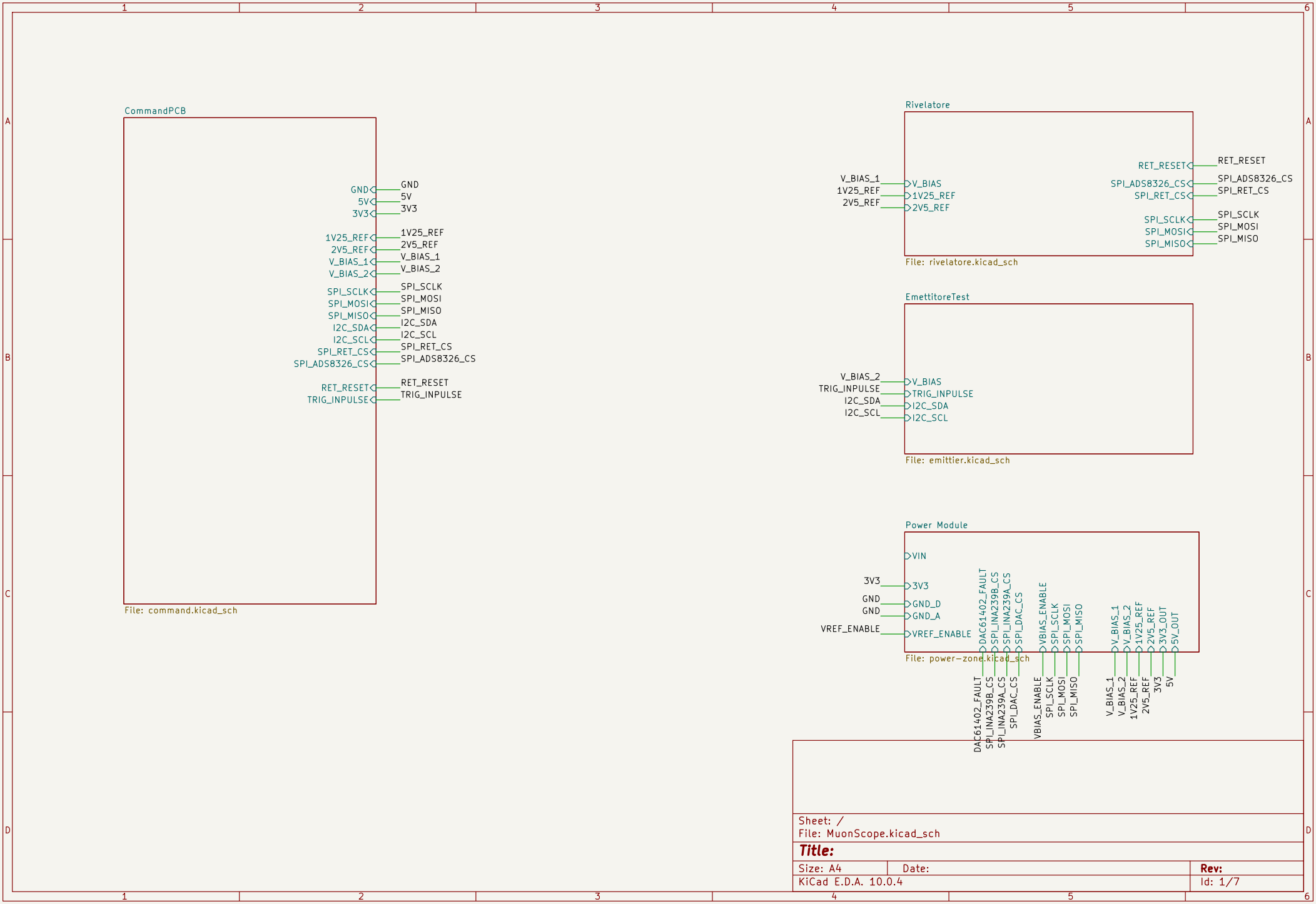
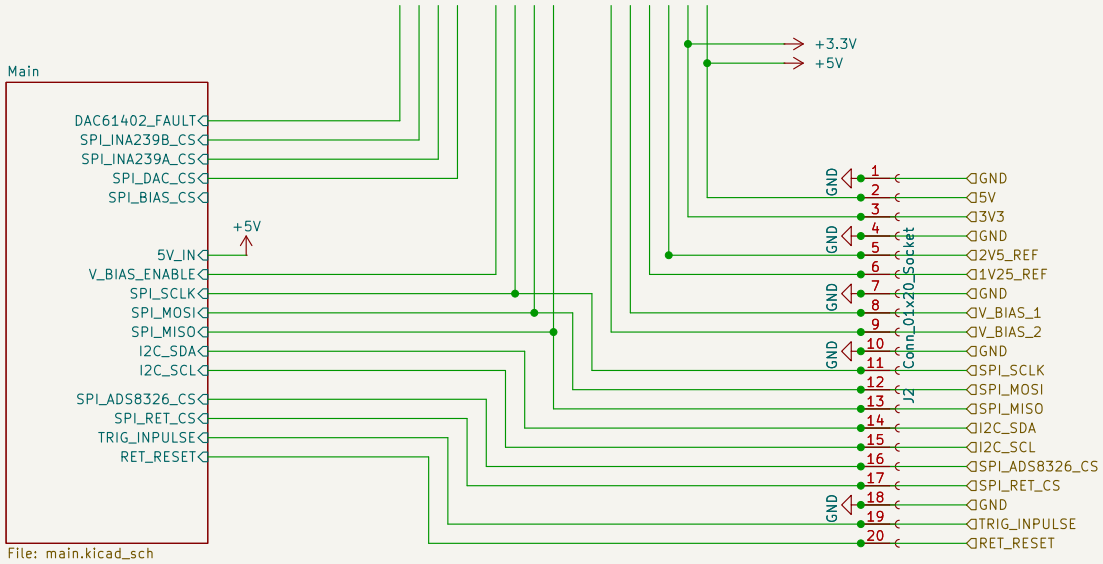
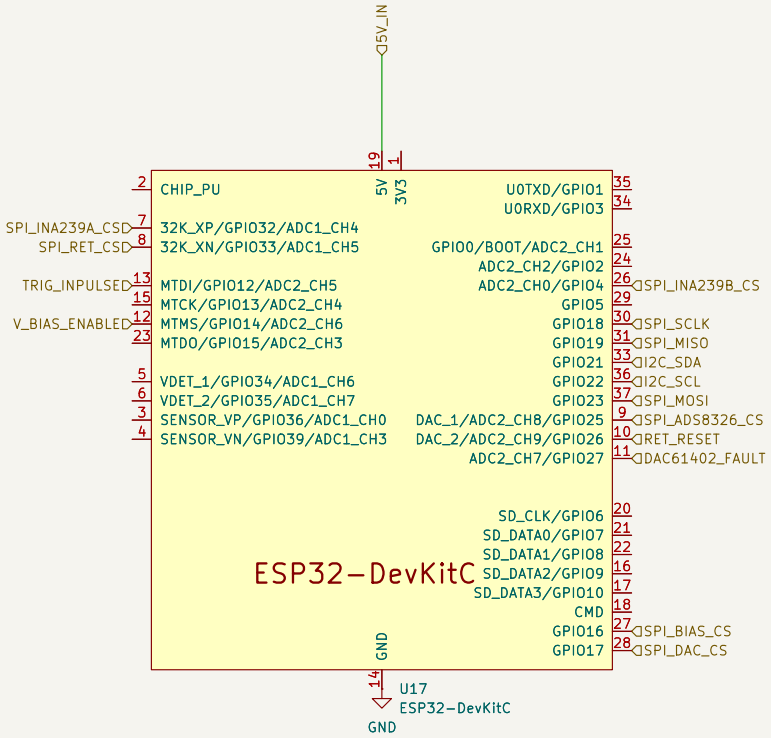

CommandPCB
GND
5V
3V3
1V25_REF
2V5_REF
V_BIAS_1
V_BIAS_2
SPI_SCLK
SPI_MOSI
SPI_MISO
I2C_SDA
I2C_SCL
SPI_RET_CS
SPI_ADS8326_CS
RET_RESET
TRIG_INPULSE
File: command.kicad_sch
Rivelatore
V_BIAS_1
1V25_REF
2V5_REF
V_BIAS
1V25_REF
2V5_REF
RET_RESET
SPI_ADS8326_CS
SPI_RET_CS
SPI_SCLK
SPI_MOSI
SPI_MISO
File: rivelatore.kicad_sch
EmettitoreTest
V_BIAS_2
TRIG_INPULSE
I2C_SDA
I2C_SCL
V_BIAS
TRIG_INPULSE
I2C_SDA
I2C_SCL
File: emittier.kicad_sch
Power Module
3V3
GND
GND
VREF_ENABLE
VIN
3V3
GND_D
GND_A
VREF_ENABLE
DAC61402_FAULT
SPI_INA239B_CS
SPI_INA239A_CS
SPI_DAC_CS
VBIAS_ENABLE
SPI_SCLK
SPI_MOSI
SPI_MISO
V_BIAS_1
V_BIAS_2
1V25_REF
2V5_REF
3V3_OUT
5V_OUT
File: power-zone.kicad_sch
Sheet: /
File: MuonScope.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.4
Id: 1/7



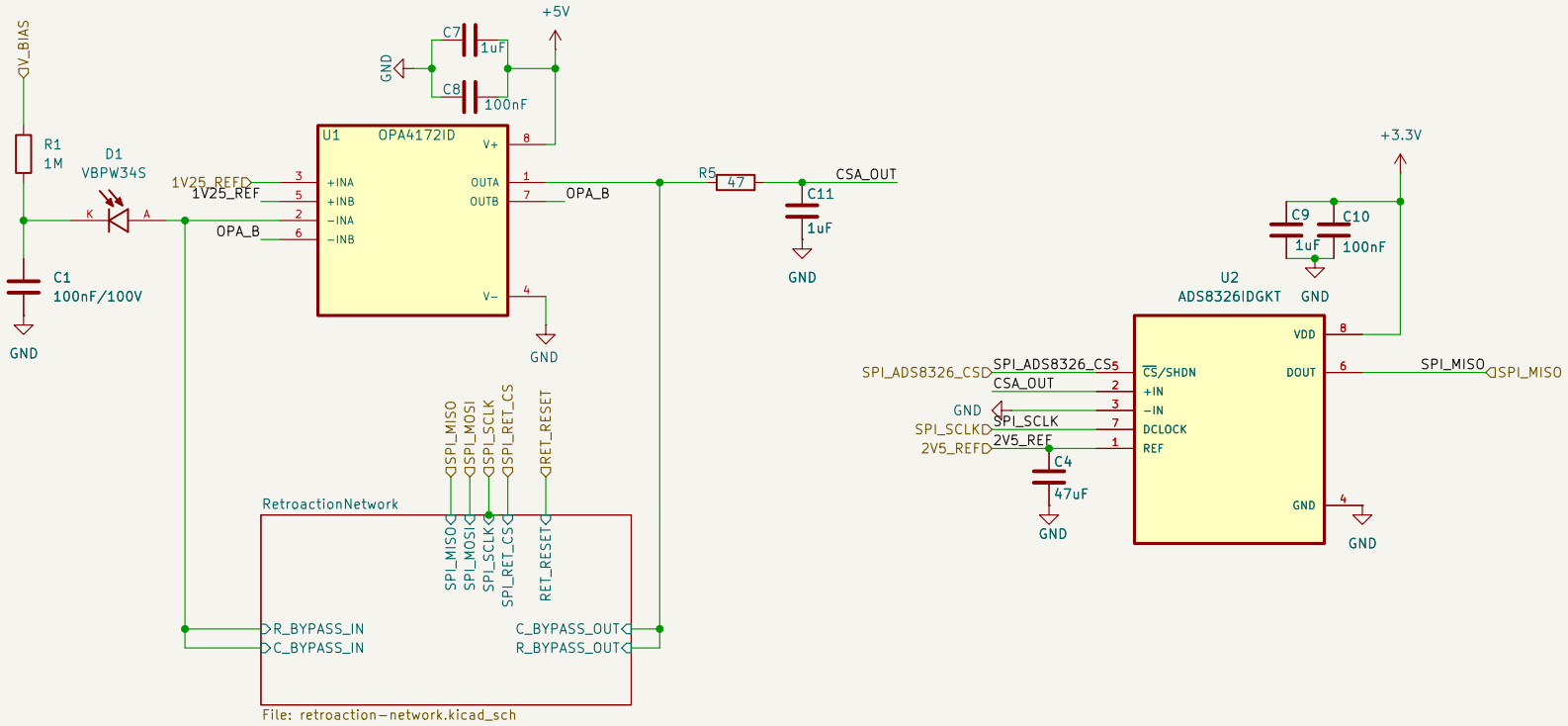
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File: command.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 3/7



Sheet: /CommandPCB/Main/
File: main.kicad_sch

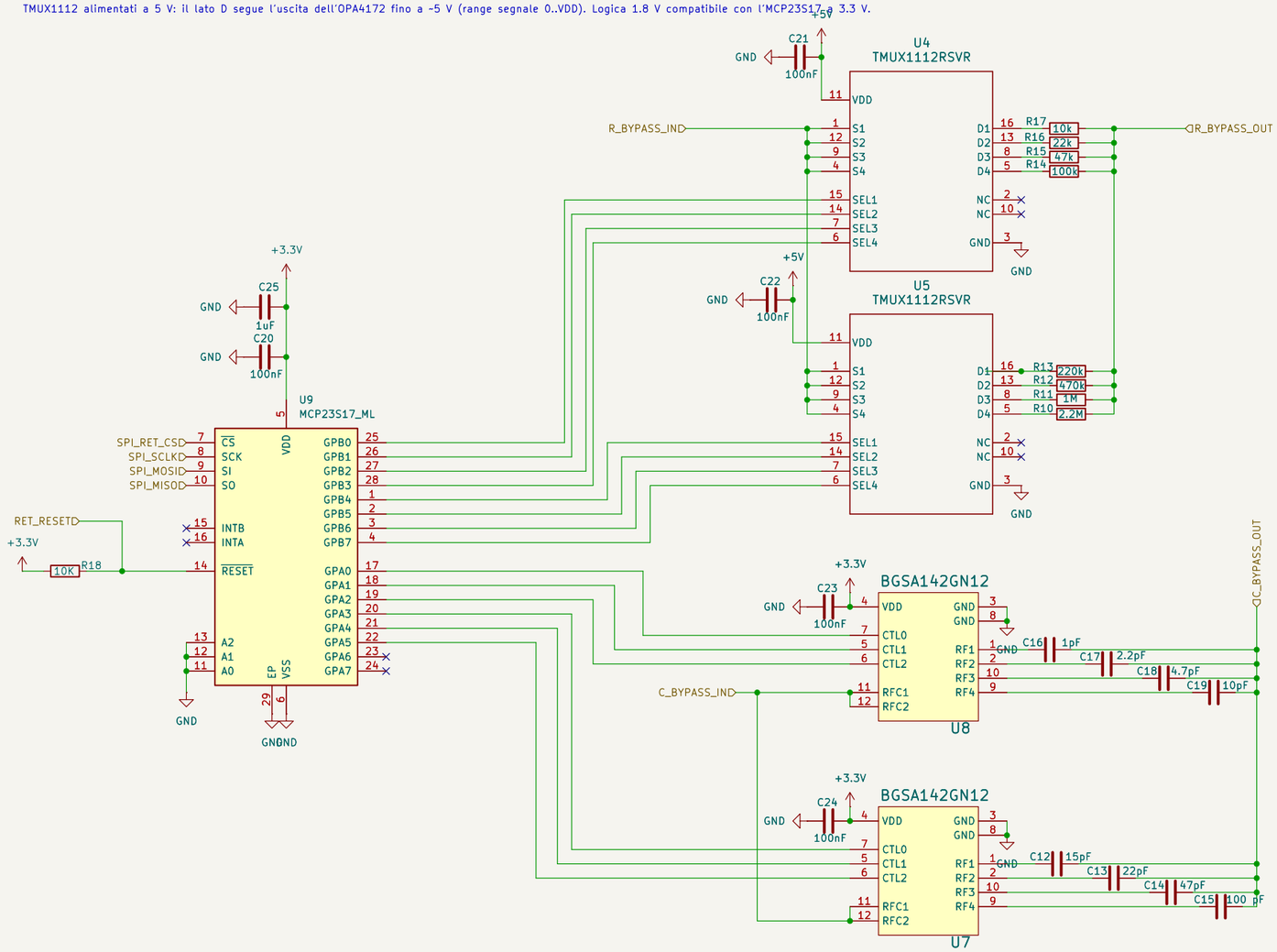
Title:

Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 6/7

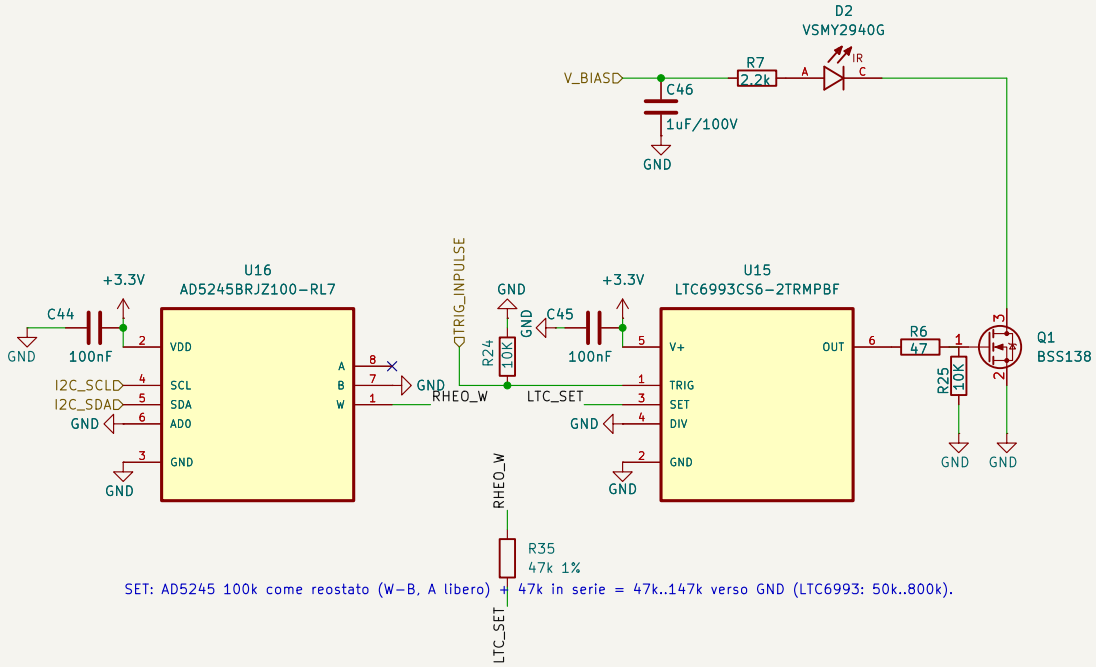


Sheet: /Rivelatore/		
File: rivelatore.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 4/7

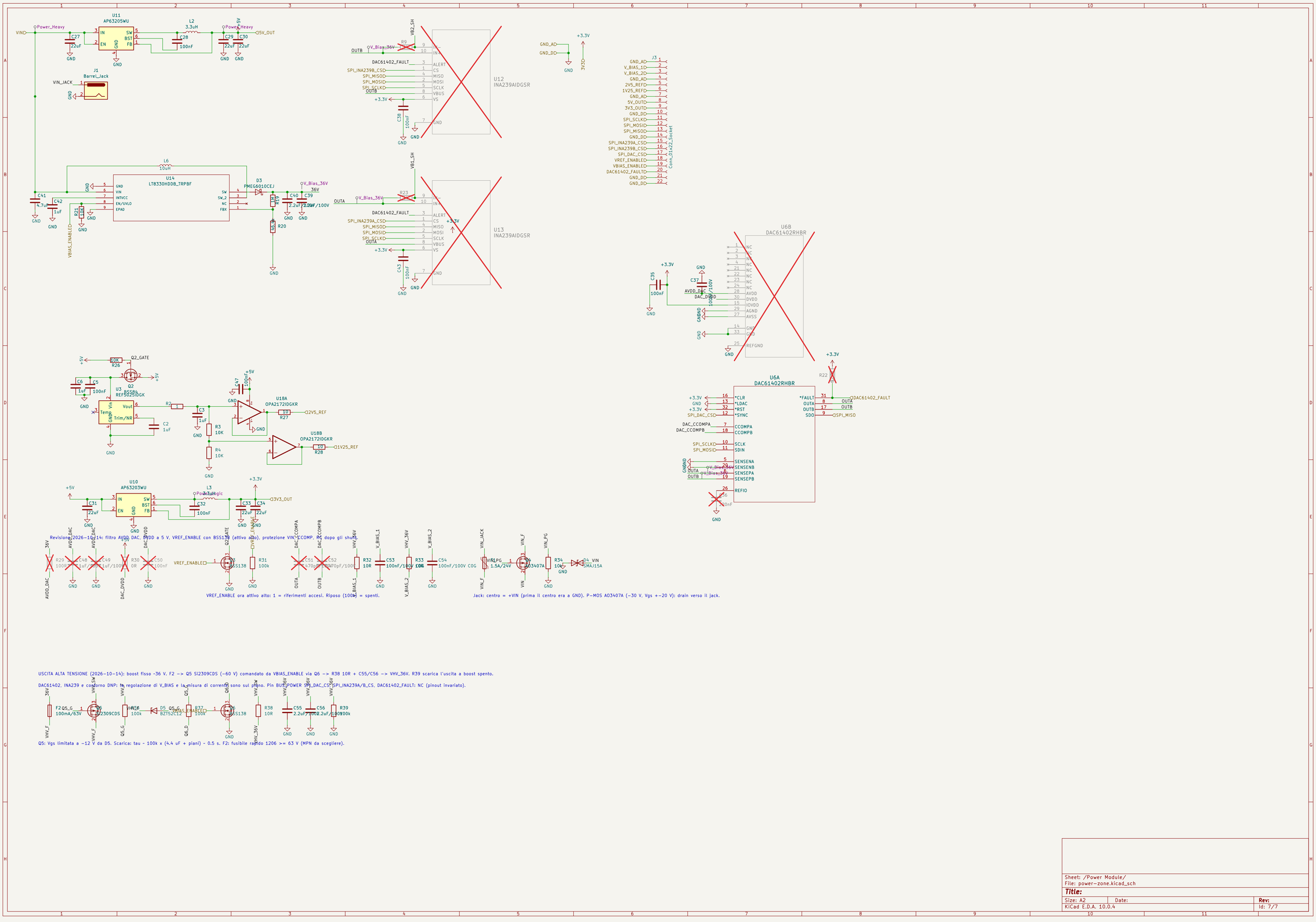
TMUX1112 alimentati a 5 V: il lato D segue l'uscita dell'OPA4172 fino a ~5 V (range segnale 0..VDD). Logica 1.8 V compatibile con l'MCP23S17 a 3.3 V.



Sheet: /Rivelatore/RetroactionNetwork/		
File: retroaction-network.kicad_sch		
Title:		
Size: A3	Date:	**Rev:**
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Sheet: /EmettitoreTest/		
File: emittier.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.4		Id: 5/7



Revisione 2026-10-14: filtro AVDD DAC, DVDD a 5 V, VREF_ENABLE con BSS138 (attivo alto), protezione VIN_CCOMP. R24 dopo gli shunt.

VREF_ENABLE ora attivo alto: 1 = riferimenti accesi. Riposo (100k) = spenti.

Jack: centro = +VIN (prima il centro era a GND). P-MOS AO3407A (−30 V, Vgs +−20 V): drain verso il jack.

USCITA ALTA TENSIONE (2026-10-14): boost fisso -36 V. F2 -> Q5 SI2309CDS (−60 V) comandato da VBIAS_ENABLE via Q6 -> R38 10R + C55/C56 -> VHV_36V. R39 scarica l'uscita a boost spento.

DAC61402, INA239 e connettore DNP: la regolazione di V_BIAS e la misura di corrente sono sul piano. Pin BUS_POWER su I_DAC_CS, SPI_INA239A/B_CS, DAC61402_FAULT: NC (pinout invariato).

Q5: Vgs limitata a −12 V da D5. Scarica: tau ≈ 100k x (4.4 uF + piani) ≈ 0.5 s. F2: fusibile rapido 1206 >= 63 V (MPN da scegliere).

Sheet: /Power Module/
File: power-zone.kicad_sch

Title:

Size: A2 | Date: | **Rev:**
KiCad E.D.A. 10.0.4 | Id: 7/7